

Subject Description Form

Subject Code	AF4615
Subject Title	Financial Economics
Credit Value	3
Level	4
Normal Duration	One Semester
Pre-requisite / Co-requisite/ Exclusion	Mathematics Pre-requisite: AMA1500 Foundation Mathematics for Accounting and Finance or AMA1501 Introduction to Statistics for Business or AMA1100 Basic Mathematics or AMA1131 Calculus or equivalent; <u>AND</u> Finance Pre-requisite: AF1605/AF1BN02 Introduction to Economics or AF3625 Engineering Economics or AF3313 Business Finance or AF4320 Corporate Finance or equivalent
Objectives	There are two major emphases in this course. The first part of the course introduces key quantitative methods for financial analysis. The second part of the course focuses on economic theories that form the fundamentals in financial economics.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Develop ability to apply economic theories that underlie the foundations of modern financial economics. Apply critical thinking in different business contexts and demonstrate creative thinking in idea expression and presentation. b. Analyze and solve real world financial economic issues related to risk and asset pricing. c. Understand the “the three pillars” of finance, and apply basic financial theories and understand the operations of financial markets. d. Get prepared for postgraduate-level studies.

Subject Synopsis/ Indicative Syllabus	Economic Foundations and the Three Pillars of Finance Utility; General Equilibrium; Competitive Equilibrium; Financial Markets and Social Welfare; Time Value of Money Discounting; Yield Curve; Term Structure of Interest Rate; Spread; Bond Pricing Uncertainty and Risk Risk and Uncertainty; Preference Theory; Choice under Uncertainty; Expected Utility; von Neumann-Morgenstern Utility; Absolute and Relative Risk Aversion; Risk Premium and Certainty Equivalence; Stochastic Dominance; Mean Preserving Spread; Risk Aversion and Risky Portfolio Composition; Separating Risk and Time Preferences; Systematic Risk; Idiosyncratic Risk; CAPM Optimization over Time: Resource Allocation over Time under Conditions of Uncertainty Dynamic Control; Bellman Equation; Lucas Tree Model; CCAPM Financial Institutions Bank Balance Sheet; Asset Transformation; Liquidity Management; Asset Management; Liability Management; Capital Adequacy Management; Liquidity and Solvency; DD Model; Financial Disintermediation; Financial Structure
Teaching/Learning Methodology	Lectures will be given in the first part of the teaching session, to give structure to the development of the subject, and to explain central concepts and ideas. The rest of the class time will be allocated to students' presentations and problem solving.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d		
	1. Problem Set	20%	✓	✓	✓	✓		
	2. Presentation and Report	25%	✓	✓	✓	✓		
	3. Participation	5%	✓	✓	✓	✓		
	4. Final examination	50%	✓	✓	✓	✓		
	Total	100 %						
	*Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Problem set –Problem set questions are used to test students’ ability in achieving the intended learning outcomes through a more in-depth investigation of different topics. Presentation and report – Students are expected to apply financial economic theories to analyze investment and corporate finance issues. Class participation – Students should read assigned readings before the class to prepare for better learning and possible Q&A sessions in class. Final examination is used to test students’ overall ability in achieving the intended learning outcomes. Note: The specific requirements on individual assessment components discussed above could be adjusted based on the pedagogical needs of subject lecturers.							
Student Study Effort Expected	Class contact:							
	▪ Lectures / Seminars				39 Hrs.			
	Other student study effort:							
	▪ Preparation for presentation and examination				50 Hrs.			
	▪ Self study and homework preparation				50 Hrs.			
	Total student study effort				139 Hrs.			

Reading List and References	<u>Textbook</u> Bodie, Z., Merton, R.C. and Cleeton, D.L., 2009. <i>Financial Economics</i>, 2nd, Pearson College Div <u>Reference</u> Bodie, Z., Kane, A. and Marcus, A.J., 2023. <i>Investments</i>, 13th, McGraw-Hill. Cochrane, J.H., 2009. <i>Asset Pricing: Revised Edition</i>. Princeton university press. Degryse, H., Kim, M. and Ongena, S., 2009. <i>Microeconometrics of Banking: Methods, Applications, and Results</i>. Oxford University Press. Freixas, X. and Rochet, J.C., 2023. <i>Microeconomics of Banking</i>, 3rd, MIT press. Ross, S., Westerfield, R., Jaffe, J. and Jordan, B., 2022. <i>Corporate Finance</i>, 13th, McGraw Hill. Varian Hal R, 2024. <i>Intermediate Microeconomics: a Modern Approach</i>, 10th, W. W. Norton & Company.
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